CoDMAV

PES UNIVERSITY

Summer Internship, June-July,2025- CODMAV
Review-2

Project Title : Credit risk analysis using PQC FL
Project ID : 10
Project Guide : Dr P Kavitha
Project Team Members : Chandana Lakya [PES2UG23CS143]
Pranav Manoj [PES2UG23CS428]
Deliverable Submitted: IEEE paper
Software Prototype

Outline

- Abstract
- Problem Statement and Research Gap
- Literature Survey(At least 5 recent papers)
- Suggestions from review-2
- Objectives
- Design(Architecture/Flowchart/use case diagram/or Class Diagram)
- Methodology Adopted- Algorithms/Techniques Adopted
- Implementation- Code snippets
- Results – output screenshots
- Conclusion and Future Enhancement
- References

Abstract

Federated learning offers a compelling solution by enabling multiple financial institutions to collaboratively train machine learning models without sharing raw data, thus preserving privacy while leveraging distributed datasets for improved model performance.
Credit risk analysis requires a varying dataset base from a number of banks allowing it to truly recognise the risk factor of an applicant for a credit, and hence Federated Learning fits perfectly for the use case of banks as edge devices.

This work addresses the growing threat of quantum computing by integrating Post-Quantum Cryptography (PQC)—specifically lattice-based and code-based approaches—into the FL process for credit risk analysis. By securing communication and model updates with quantum-resistant algorithms, our framework ensures long-term data privacy and system integrity. The significance of this research lies in creating a secure, scalable, and future-proof FL infrastructure for the financial industry, bridging the gap between privacy-preserving analytics and quantum-resilient security.

Problem Statement and Research Gaps Identified

Our Problem statement dives deep into addressing the privacy issues faced by fields dealing in sensitive data for example banking to incorporate AI driven solutions without having to compromise on the security threats faced when implementing a traditional centralised repository of data.
We've chosen the topic of Credit risk Analysis using a Post Quantum Secure Federated Learning model , to show the effectiveness and robustness this model has by integrating the future of secure communications using NIST approved PQC algorithms for encryption of weights communication between the aggregation server and clients.

Literature Survey

Approach	Limitations
Standard Federated Learning	Vulnerable to quantum attacks due to use of classical cryptography (RSA, ECC)
FL with Homomorphic Encryption	High computational and memory overhead; not post-quantum secure
Blockchain-based PQC-FL	Focus is on authentication, not model privacy or financial applications
Signature-Centric PQC in FL	Limited scalability testing; does not address credit risk modeling or resource allocation

Objectives

The objectives that we set out are the following :

- Successfully integrate a working PQC wrapper over a widely known FL framework (Flwr)
- Enable Asymmetric communication protocols for improvements in resource management.
- Plot metrics and research on the tradeoffs between security and resource allocation via PQC.

Methodology Adopted

To effectively see the tradeoffs that we have hypothesized between security and a multitude of other features , we have developed a few cases that start from an ideal pure FL to the realistic scenario that we hope to address. Here are few of the scenarios :

1. Pure FL with IID data
2. FL - IID Data with PQC overhead
3. FL with Non IID data with Classical schemes
4. FL - NIID data with Asymmetric PQC

These four cases that we decided to go over shows first in the ideal scenario what the upper bound on resource management, accuracy and aggregation quality would look for our experiment , and as per our hypothesis this would vary in multiple ways with each offering its own pros and cons.

Design

So diving first into the FL setup , we have sourced a public dataset from HuggingFace relating to Credit data in an Indian Environment having multiple features like Gender, Marital status, income and occupation.
We will be using the FL framework flwr to enable our FL environment to learn from a test case of 20 users across 20 clients, each client having skewed and disproportionate data to stimulate non IID scenarios.
To enable PQC algorithms using python we are using the open sourced library Liboqs-python and have implemented a wrapper logic for it to be reproducible and robust across all versions.
The metrics that we would be logging are Accuracy,Communication overhead,Training time,Aggregation quality, and Resource usage.
We then plan on implementing the asymmetric approach and log the same metrics over

Implementation

1. **Initialize Global Model**
   The central server creates an initial machine learning model.

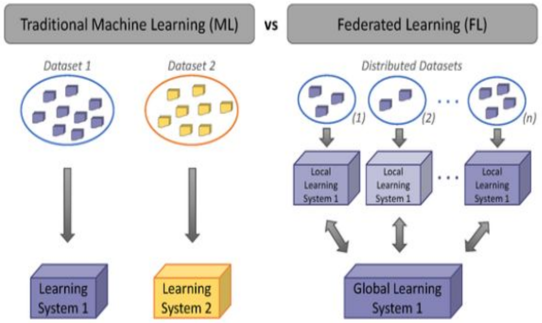
2. **Distribute Model to Clients**
   The server sends the global model to selected client devices.

3. **Local Training on Clients**
   Each client trains the model on its private data.
   No raw data is shared.

4. **Send Model Updates to Server**
   Clients send only weight updates or gradients.

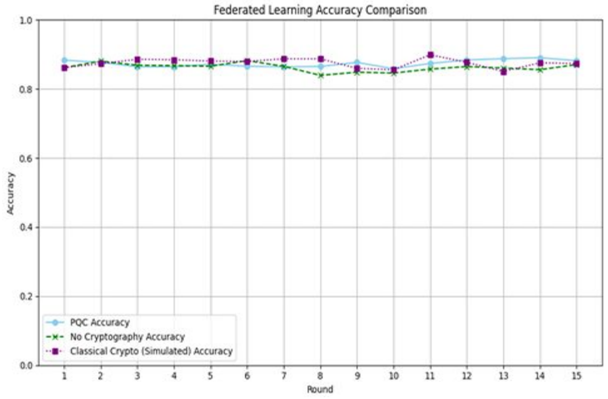
5. **Aggregate Updates at Server**
   Server uses algorithms like **Aggregation** to combine updates and improve the global model.

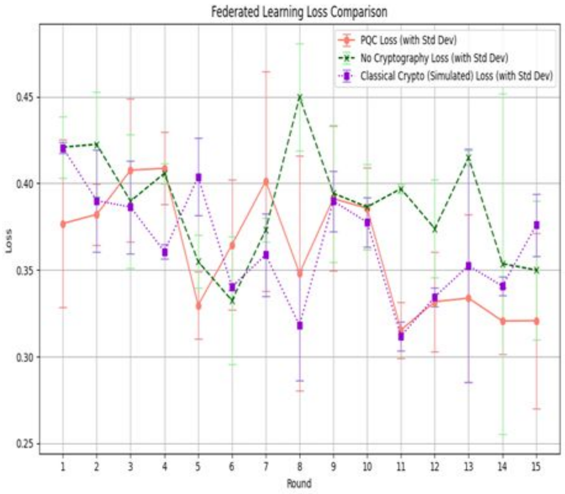
6. **Repeat Rounds**
   Steps 2–5 are repeated until the model reaches acceptable performance.

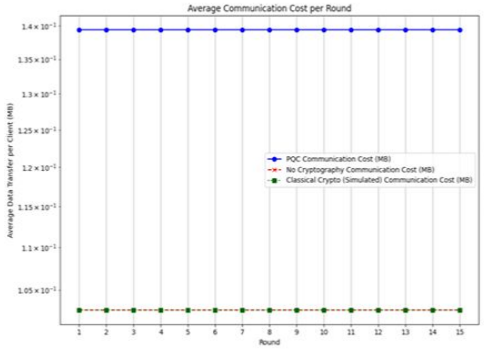


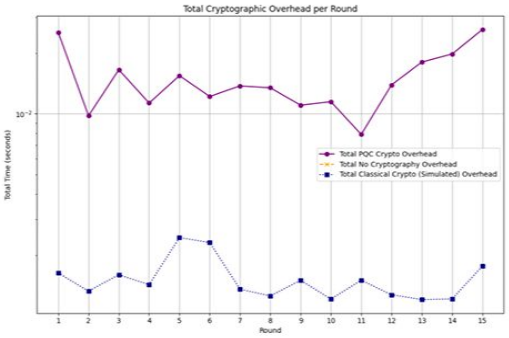
CoDMAV

Results









Conclusion and Future Enhancement

Operational feasibility

Our primary finding is that the PQC FL integration is successful and stable and we have presented it as a POW prototype on our github. This ran for 20 rounds for 20 clients without errors or model corruption.

Security Posture
We have successfully elevated the systems security from being vulnerable to quantum attacks and specifically against threats like shor's algorithm.

Final thesis statement
We have demonstrated the hypothesized tradeoffs between security with accuracy and other parameters.
While incurring a predictable performance cost of , our implementations using kyber512 maintains complete model integrity, providing a clear and practical pathway for building secure, quantum ready systems.

CoDMAV

PES UNIVERSITY

References

Wang, Zhaoyang, Yuchen Zhang, Yue Wang, and Kun Xie. "FedCredit: A Federated Learning Framework for Credit Risk Assessment." *2022 International Conference on Big Data, Information and Computer Network (BDICN)* (2022).

Ma, X., H. Jiang, J. Wang, and Y. Zhang. "A deep learning-based federated learning approach for credit scoring." *Soft Computing* (2023).

Commey, Daniel, and Garth V. Crosby. "PQS-BFL: A Post-Quantum Secure Blockchain-based Federated Learning Framework." *arXiv preprint arXiv:2505.01866* (2025).

Gharavi, Hadi, Jorge Granjal, and Edmundo Monteiro. "PQBFL: A Post-Quantum Blockchain-based Protocol for Federated Learning." *arXiv preprint arXiv:2502.14464* (2025).

Li, Pingzhi, Tianlong Chen, and Junyu Liu. "Enhancing Quantum Security over Federated Learning via Post-Quantum Cryptography." *2024 IEEE 6th International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)*. IEEE, 2024.

Thank
You